

Project Report: Data Visualization Using Power BI

Project Title: Data Visualization on an E-commerce Dataset

Internship: Decode Labs Internship – Project 4

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Introduction

This project focused on creating an interactive dashboard using Microsoft Power BI. The aim was to present e-commerce sales data in a way that makes it easy to understand business performance and identify useful insights. Rather than viewing hundreds of rows of data, the dashboard summarizes the information into visuals that support quick decision-making.

The dataset used for this project is the same e-commerce dataset that was analyzed during the previous SQL project. Since the data had already been cleaned, the main focus of this project was on designing meaningful visualizations and presenting the results clearly.

Project Objective

The objective of this project was to:

- Import the cleaned dataset into Power BI.
- Create important business KPIs.
- Build interactive charts and tables.
- Allow users to filter the dashboard using slicers.
- Present business insights in a simple and professional dashboard.

Dataset Description

The dataset contains sales transactions from an e-commerce business.

Dataset Summary:

- Total Records: 1,200
- Total Columns: 14

The dataset includes information such as:

- Order ID
- Product

- Quantity
- Total Price
- Payment Method
- Order Status
- Referral Source
- Customer Information

Data Preparation

Before building the dashboard, the dataset was imported into Power BI.

The following preparation steps were completed:

1. Imported the cleaned CSV dataset into Power BI.
2. Verified that all columns had the correct data types.
3. Formatted the Total Price field as Currency.
4. Created measures to calculate Total Sales, Total Orders, and Average Order Value.
5. Checked that the visuals displayed accurate values.

Since the dataset had already been cleaned in previous projects, no additional data cleaning was required.

Dashboard Development

The dashboard was designed to provide a quick overview of sales performance.

The following components were included:

- KPI Cards showing Total Sales, Total Orders, and Average Order Value.
- Bar Chart displaying Total Sales by Product.
- Column Chart showing Sales by Payment Method.
- Column Chart showing Average Order Value by Product.
- Donut Chart illustrating the distribution of Order Status.
- Bar Chart displaying Sales by Referral Source.
- Product Sales Summary table.
- Interactive slicers for Product, Payment Method, and Order Status.

These visuals allow users to interact with the dashboard and explore different aspects of the sales data.

Key Findings

The dashboard revealed several important insights:

- The business generated approximately \$1.26 million in total sales.
- A total of 1,200 orders were recorded.
- The average order value was approximately \$1,053.97.
- Chair and Printer generated the highest sales revenue among the available products.
- Credit Card and Online payments contributed the highest sales.
- Instagram and Email were the strongest referral sources.
- Orders were distributed fairly evenly across Delivered, Shipped, Pending, Returned, and Cancelled statuses.

These insights can help management understand customer purchasing patterns and identify areas for improvement.

Challenges Encountered

One challenge encountered during the project was ensuring that sales values were displayed correctly as currency. Proper formatting was applied so that all monetary values appeared consistently across the dashboard.

Another challenge involved selecting the most suitable visuals for each type of analysis. Different chart types were evaluated before choosing those that best represented the data.

Conclusion

This project demonstrated how Power BI can transform raw sales data into meaningful business insights through interactive dashboards. By combining KPIs, charts, tables, and filters, the dashboard provides a clear picture of sales performance and customer activity.

Completing this project strengthened my understanding of data visualization, dashboard design, and the importance of presenting analytical results in a way that is both informative and easy to interpret.